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An Empirical Analysis of Analysts' Target Prices: Short-Term Informativeness and Long-Term Dynamics

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1 Big picture

1.1 Question

The paper asks whether sell-side analysts' target prices contain information for equity prices. The central empirical question is not merely whether analysts issue target prices, but whether investors treat revisions in those target prices as informative beyond two other analyst signals: stock recommendations and earnings forecast revisions.

A second question is dynamic rather than event-based: how do consensus target prices and market prices move together over longer horizons? Since target prices are normally one-year-ahead forecasts, the target-price-to-market-price ratio can be interpreted as an analyst-implied expected return. The paper therefore studies both short-run informativeness and long-run adjustment.

1.2 Setting

The sample uses First Call analyst target prices, stock recommendations, and earnings forecasts from 1997 through 1999. The authors restrict the main event tests to reports coded as "real time," which is important because the paper's central evidence uses announcement-window abnormal returns.

The target-price database is large: 223,016 target price reports for 6,544 distinct firms over 1997–1999. These firms account for a very large share of CRSP market value, so the analysis is not focused only on obscure or micro-cap firms. The authors also use 117,114 recommendations and 124,286 earnings forecasts.

1.3 Main findings

- **Target price revisions are informative.** Abnormal returns around target price announcements increase monotonically with the favorableness of target price revisions.
- **Target prices add information beyond recommendations and earnings forecasts.** In event regressions, the target price revision coefficient remains large and highly significant after controlling for recommendation upgrades/downgrades/reiterations and earnings forecast revisions.
- **The market reaction is strongest when signals reinforce each other.** Recommendation upgrades with positive target price revisions and downgrades with negative target price revisions generate especially strong announcement reactions.
- **Target price revisions predict future returns.** Post-event abnormal returns are positive after favorable target price revisions, especially for recommendation upgrades and reiterations.
- **Target and market prices are cointegrated.** The average long-run target-price-to-market-price ratio is about 1.28, meaning that target prices are on average 28% above market prices.
- **Analysts, not prices, do most of the long-run correction.** When target and market prices move away from their long-run relation, the consensus target price adjusts back much more than the market price does.

1.4 Economic takeaway

Target prices are not merely cosmetic labels attached to recommendations. They summarize analysts' valuation magnitudes, likely through valuation multiples, and market participants respond to these magnitudes. The paper shows that recommendations answer a coarse directional question - buy, hold, or sell - while target prices communicate the size and valuation content of the analyst's belief.

2 Incremental contribution of the paper

2.1 Relative to the analyst literature

Earlier empirical work mostly focused on recommendations and earnings forecasts. This paper adds the third major analyst output: target prices. It shows that target prices have both immediate information content and predictive content for post-announcement returns.

The contribution is incremental because recommendations are discrete and coarse. A recommendation reiteration can still contain new information if the analyst revises the target price substantially. The paper shows that this is exactly what happens: recommendation reiterations have meaningful announcement returns when accompanied by large target price changes.

2.2 Relative to valuation and price-formation research

Target prices are often produced from a reduced-form valuation exercise, such as forecasted earnings times a multiple. When target price revisions remain informative after controlling for earnings forecast revisions, the residual information is plausibly information about the multiple, the analyst's valuation model, or the analyst's confidence in the firm's prospects.

This makes target prices useful for studying how investors process analysts' valuation views, not simply analysts' earnings forecasts.

2.3 Relative to long-run expected-return measurement

The cointegration analysis treats the target-price-to-market-price ratio as an analyst-implied expected return. This provides a market-based and analyst-based ex ante expected return measure, rather than relying only on realized returns or asset-pricing model estimates.

The paper therefore links analyst research, event-study information content, and asset-pricing dynamics in one design.

3 Data, sample construction, and variables

3.1 First Call data

The paper combines three First Call databases:

- Target prices: 223,016 target price reports, 6,544 firms, 190 brokerage houses.

- Stock recommendations: 117,114 reports, 8,673 firms, 325 brokerage houses.
- Earnings forecasts: 124,286 reports, 9,167 firms, 282 brokerage houses.

The sample period is 1997–1999, and the authors emphasize accurate dating by using only real-time observations in their event tests.

3.2 Target price information measures

The paper constructs two main measures.

Level of target price relative to price.

$$\frac{TP}{P} = \frac{\text{announced target price}}{\text{stock price two days before the announcement}}.$$

Since most target prices are one-year-ahead target prices, this ratio can be interpreted as an analyst-implied expected gross return. For example, a ratio of 1.30 means the analyst target is 30% above the pre-announcement stock price.

Revision in target price relative to price.

$$\frac{\Delta TP}{P} = \frac{\text{current target price} - \text{same broker's prior target price}}{\text{stock price two days before the announcement}}.$$

This is the main event-study measure because it captures new information relative to the same brokerage house's previous target.

The paper also reports

$$\frac{\Delta TP}{TP_{-1}},$$

which is the percentage revision in the brokerage house's target price.

3.3 Recommendation and earnings forecast variables

Recommendations are categorized as upgrades, downgrades, or reiterations. Sell and strong sell recommendations are combined because they are relatively rare.

The earnings forecast revision is

$$\frac{\Delta F}{P},$$

where ΔF is the change in the brokerage house's current fiscal-year earnings forecast, scaled by the stock price two days before the announcement.

3.4 Abnormal returns

For announcement-window tests, abnormal returns are buy-and-hold stock returns minus the buy-and-hold NYSE/AMEX/Nasdaq value-weighted market return over days $[-2, +2]$ around the target price announcement.

For post-event tests, the paper computes size- and book-to-market-adjusted cumulative abnormal returns over months +1 through +6. It also uses calendar-time portfolio regressions with the Carhart four-factor model.

4 Short-run informativeness of target prices

4.1 Unconditional event evidence

The paper first ranks target price announcements into deciles using TP/P and $\Delta TP/P$. The main pattern is monotonic: higher target price levels and revisions are associated with higher announcement-window abnormal returns.

The strongest spread comes from $\Delta TP/P$. The least favorable revision portfolio has an average abnormal return of about -3.96% , while the most favorable has about $+3.21\%$. This is the first evidence that investors do not ignore target price magnitudes.

4.2 Regression design

The core event regression is

$$AR = \alpha_1 UPGRADES + \alpha_2 DOWNGRADES + \alpha_3 REITERATIONS + \beta \left(\frac{\Delta F}{P} \right) + \gamma \left(\frac{\Delta TP}{P} \right) + \varepsilon.$$

Here AR is the market-adjusted announcement return. The coefficient γ measures whether target price revisions are incrementally informative after controlling for recommendations and earnings forecasts.

4.3 Main regression result

In the full sample, the target price revision coefficient is 3.705 with a t -statistic of 38.7. The paper interprets this as economically large: a one-standard-deviation increase in $\Delta TP/P$ raises the event-day abnormal return by about 2.9 percentage points.

Recommendations and earnings forecast revisions also remain informative, which means none of the three signals completely subsumes the others.

4.4 When is a target price revision especially informative?

Target prices are more informative in some states:

- **Recommendation downgrades.** The target price coefficient is particularly high when the target revision comes with a recommendation downgrade. This fits the idea that unfavorable analyst signals are more credible because analysts are reluctant to issue bad news.
- **Consistent signals.** When recommendation direction and target price direction reinforce each other, announcement returns are stronger.
- **Recommendation reiterations.** A reiteration is not necessarily uninformative. If the target price changes meaningfully, investors react even when the buy/hold/sell label stays unchanged.

5 Post-event abnormal returns and underreaction

5.1 Event-time evidence

The paper next asks whether target price revisions predict returns after the announcement window. The most important evidence appears for recommendation upgrades and reiterations.

For upgrades, the average six-month CAR is 3.08% for all target price revisions and 5.21% for the most favorable target price revisions. For reiterations, the highest target price revision group has a six-month CAR of 6.22%, while the lowest target price revision group has a six-month CAR of -1.88%.

Recommendation downgrades show much less drift in the post-event window.

5.2 Calendar-time evidence

Because long-run event studies can be sensitive to cross-sectional dependence and compounding, the paper also forms calendar-time portfolios and runs Carhart four-factor regressions:

$$r_{p,t} - r_{f,t} = \alpha + \beta_1 RMRF_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 PR12_t + \epsilon_t.$$

The portfolio is formed from firms with target price announcements in the previous six months. Evidence of abnormal performance comes from the intercept α .

The calendar-time results confirm the event-time pattern for favorable target price revisions. For recommendation upgrades, the most favorable target price revision portfolio has a monthly alpha of 0.799% with a t -statistic of 2.59. For recommendation reiterations, the most favorable target revision portfolio has a monthly alpha of 0.478% with a t -statistic of 3.07.

5.3 Interpretation

The post-event evidence is consistent with underreaction or rational learning under uncertainty. The authors are careful because the sample period, 1997–1999, overlaps a historically unusual bull market. Still, within that sample, target price revisions clearly contain information for subsequent returns.

6 Long-run dynamics between target prices and market prices

6.1 Why cointegration?

Market prices and target prices are both nonstationary time series. However, if analysts are forecasting the same underlying value that the stock market is pricing, the two price series should share a long-run relation.

The paper studies the target-price-to-market-price ratio as this long-run relation. If target prices are one-year-ahead forecasts, then

$$\frac{TP}{P}$$

can be interpreted as the analyst-implied expected gross return.

6.2 Error-correction model

Let

$$x_t = \begin{pmatrix} tp_t \\ p_t \end{pmatrix},$$

where tp_t is the consensus target price and p_t is the market price. The paper estimates an error-correction model:

$$\Delta x_t = \sum_{i=1}^p \Gamma_i \Delta x_{t-i} + \Pi x_{t-1} + \epsilon_t.$$

The long-run impact matrix is parameterized as

$$\Pi = \alpha \beta',$$

where

$$\beta = \begin{pmatrix} -1 \\ b \end{pmatrix}.$$

The coefficient b captures the long-run target-price-to-market-price ratio. If $b = 1.28$, the long-run target price is 28% above the market price.

The vector α contains response coefficients. These tell us whether target prices or market prices do the correcting after the system deviates from the long-run relation.

6.3 IBM example

For IBM, the long-run target-to-market ratio is estimated at 1.23. The target price response coefficient is 0.17, while the market price response is statistically insignificant. This means that when IBM's target and market prices deviate from their long-run relation, analysts revise target prices back toward the relation much more than the market price adjusts.

6.4 Full-sample evidence

Across 900 firms with continuous weekly target and market price data:

- The mean long-run target-price-to-market-price ratio is 1.28; the median is 1.26.
- The target price response coefficient averages 0.09.
- The market price response coefficient averages only -0.02 .

The long-run ratio declines with firm size: small firms have an average ratio of 1.37, medium firms 1.29, and large firms 1.23. This is consistent with higher expected returns or higher perceived risk for smaller firms.

7 Connecting the short-run and long-run evidence

7.1 Expected versus unexpected target price revisions

The paper then asks whether investors understand the long-run dynamics. If investors anticipate some predictable target price revisions, only the unexpected component should move prices.

For each event, the authors estimate an expected target price revision from the cointegration model and then define the unexpected revision as actual minus expected. The regression is

$$AR = \alpha_1 UPGRADES + \alpha_2 DOWNGRADES + \alpha_3 REITERATIONS + \beta \left(\frac{\Delta F}{P} \right) + \gamma \left(\frac{Expected \Delta TP}{P} \right) + \eta \left(\frac{Unexpected \Delta TP}{P} \right) + \varepsilon.$$

7.2 Main result

The expected target price revision is not reliably related to announcement returns. The unexpected target price revision is strongly related to returns, with a coefficient of 2.671 and a t -statistic of 27.0.

This supports the view that investors understand the predictable component of analyst target price behavior. They react mainly to target price news that is not explained by the long-run adjustment process.

7.3 Directional asymmetry

When the target-to-market ratio is below the long-run relation and an unexpected revision moves further away from that relation, the return response is especially large in magnitude. In Table VII, the coefficient for this case is 6.864 with a t -statistic of 20.1. The market therefore reacts strongly when the target price revision is surprising and directionally meaningful.

8 Conceptual interpretation

8.1 What target prices add beyond earnings forecasts

Earnings forecasts provide one input to valuation, but target prices combine earnings forecasts with a valuation multiple, a required return, or other analyst judgments. Therefore, target prices can contain information about the analyst's implied valuation model even after earnings forecast revisions are controlled for.

8.2 What target prices add beyond recommendations

Recommendations are categorical. A buy recommendation can cover a wide range of expected upside. A target price tells investors whether the analyst expects modest upside or very large upside. This explains why target price revisions are especially useful when recommendations are reiterated.

8.3 Why analysts adjust more than prices in the long run

Market prices respond to new information at the event date. But once the target-to-market ratio deviates from its typical level, analysts appear to revise their target prices toward that long-run relation. In the paper's long-run system, analysts are the dominant error-correction force.

9 Limitations and cautions

- **Short sample window.** The 1997–1999 sample coincides with an unusual bull market. The paper explicitly notes that out-of-sample evidence would be valuable.
- **Target prices are often bundled with other analyst outputs.** The paper controls for recommendations and earnings forecast revisions, but analyst reports may contain additional text, valuation arguments, or qualitative news.
- **Event dating matters.** The authors restrict to real-time First Call reports to improve dating accuracy, but some market participants may receive analyst reports through other channels.
- **Target price horizons may be imprecise.** Most target prices are one-year-ahead forecasts, but target price meaning may vary across analysts and firms.
- **Cointegration estimates are descriptive.** The long-run target-price-to-market-price relation is estimated from data; it does not prove that analyst target prices are unbiased expected return forecasts.

10 Extracted figures and tables from the paper

10.1 Table I: Descriptive Statistics on Analysts' Target Prices, Stock Recommendations, and Earnings Forecast Revisions, 1997–1999

Descriptive Statistics on Analysts' Target Prices, Stock Recommendations, and Earnings Forecast Revisions, 1997–1999

This table reports statistics on the First Call target price (Panel A), stock recommendations (Panel B), and earnings forecasts (Panel C) databases, as well as a transition matrix of analyst stock recommendations and target prices (Panel D) for firms with available data on CRSP. To ensure accurate dating of analysts' reports, we include only observations coded as "real-time" (i.e., reports received from live feeds such as the broker note and that are dated as the date that the report was published). Panels A through C present, by year, the number of observations, the average number of reports per firm, the number of firms, the number of brokerage houses issuing reports, and the average number of brokerage houses per firm. The last row in each panel A–C presents statistics for the three-year sample period. Panel D presents the number of analyst stock recommendations (top number) and the percentage of those recommendations issued with a target price (bottom number), by changes in or reiterations of stock recommendations.

Panel A: Target Prices					
Year (1)	Price Targets		Number of Firms (4)	Brokers	
	<i>N</i> (2)	Avg. No. Per Firm (3)		<i>N</i> (5)	Avg. No. Per Firm (6)
1997	49,134	10	4,694	123	4
1998	79,936	16	4,997	136	5
1999	93,946	18	5,165	149	5
Overall	223,016	14	6,544	190	6

Panel B: Stock Recommendations					
Year	Recommendations		Number of Firms	Brokers	
	<i>N</i>	Avg. No. Per Firm		<i>N</i>	Avg. No. Per Firm
1997	32,295	6	5,572	211	4
1998	42,805	7	5,871	222	5
1999	42,014	7	5,929	210	5
Overall	117,114	6	8,673	325	7

Panel C: Earnings Forecasts					
Year	Earnings Forecasts		Number of Firms	Brokers	
	<i>N</i>	Avg. No. Per Firm		<i>N</i>	Avg. No. Per Firm
1997	39,736	6	6,474	204	5
1998	42,228	7	6,203	233	5
1999	42,322	7	6,106	246	5
Overall	124,286	7	9,167	282	7

Table I, Panels A–C: Target prices, recommendations, and earnings forecasts.

Table I (continued)

Panel D: Number of Stock Recommendations and Percentage Issued with Target Price				
From Recommendation	To Recommendation			
	Strong Buy	Buy	Hold	Sell/Strong Sell
Strong Buy	45,671 99%	5,692 52%	3,297 36%	77 39%
Buy	6,108 60%	36,823 99%	5,186 35%	114 40%
Hold	2,485 71%	4,315 70%	12,579 98%	427 42%
Sell/Strong Sell	63 56%	86 60%	424 41%	527 92%
No prior recommendation	16,374 82%	15,194 79%	8,622 49%	510 53%
Overall	70,701 91%	62,110 88%	30,108 66%	1,655 61%

Table I, Panel D: Recommendation transitions and target price issuance.

Read: The target price database contains 223,016 reports, more observations than either recommendation or earnings forecast revisions over the same period. Panel D shows that target prices are more likely to be issued with strong buy and buy recommendations than with hold or sell recommendations.

Economic message: Target prices are a major analyst output, not a rare appendix. Their uneven use across recommendation categories already suggests a selection pattern: analysts appear more willing to attach target prices to favorable opinions.

10.2 Table II: Statistics on Target Prices by Analyst Stock Recommendations

Table II
Statistics on Target Prices by Analyst Stock Recommendations

This table provides descriptive statistics on the target price information measures. Panel A provides general distributional statistics on (a) the ratio of target price to preannouncement stock price (stock price outstanding two days prior to the announcement of the target price), denoted (TP/P) , (b) the change in the individual brokerage house target price scaled by preannouncement stock price, denoted $(\Delta TP/P)$, (c) the percentage change in the brokerage house target price, denoted $(\Delta TP/TP_{-1})$, and (d) earnings forecast revision, computed as the difference in the brokerage house current and prior annual earnings forecast scaled by preannouncement stock price. Panel B provides information on the average TP/P conditional on stock recommendation revisions. Panel C reports, for each recommendation revision, averages of $\Delta TP/TP_{-1}$ as well as average price appreciation measured over the same period (since the issuance of the preceding target price). All prices and earnings are converted to the same split-adjusted basis.

Panel A: Descriptive Statistics on Measures of the Information Content of Target Price				
	Target Price to Stock Price Ratio (TP/P)	Change in Brokerage House Target Price ($\Delta TP/P$)	Change in Brokerage House Target Price ($\Delta TP/TP_{-1}$)	Forecast Revision
Mean	1.329	0.8%	5.3%	-0.41%
Max	3.004	143.3%	183.3%	35.2%
75th percentile	1.433	9.7%	8.3%	0.16%
Median	1.255	0.0%	0.0%	-0.03%
25th percentile	1.146	0.0%	-0.8%	-0.43%
Min	0.584	-136.0%	-89.0%	-422.5%
Std. Dev.	0.304	78.3%	95.9%	2.5%
<i>N</i>	204,031	115,720	115,720	82,052

Panel B: Average Target Price to Price Ratio (TP/P)				
From Recommendation	To Recommendation			
	Strong Buy	Buy	Hold	Sell/Strong Sell
Strong Buy	1.40	1.30	1.18	1.04
Buy	1.41	1.31	1.12	1.21
Hold	1.37	1.31	1.16	1.01
Sell/Strong Sell	1.42	1.36	1.11	1.03
Initiated/Resumed as	1.43	1.31	1.15	1.07
Overall	1.41	1.31	1.16	1.04

Table II, Panels A–B: Target price levels and descriptive statistics.

Table II (continued)

Panel C: Average Change in Target Price ($\Delta TP/TP_{-1}$) and Corresponding Price Appreciation

From Recommendation	To Recommendation							
	Strong Buy		Buy		Hold		Sell/Strong Sell	
Strong Buy	6.0%,	5.7%	6.4%,	2.5%	-9.9%,	-0.2%	-14.5%,	-3.5%
Buy	12.7%,	5.1%	5.4%,	4.3%	-4.5%,	2.4%	-6.1%,	5.9%
Hold	22.8%,	4.6%	16.4%,	4.4%	0.6%,	0.01%	-7.2%,	-2.9%
Sell/Strong Sell	20.6%,	1.6%	15.6%,	2.1%	11.0%,	0.7%	0.5%,	-3.5%
Initiated/Resumed as	NA		NA		NA		NA	
Overall	6.6%,	5.6%	5.7%,	4.2%	-0.4%,	0.3%	-1.7%,	-2.9

Table II, Panel C: Target price revisions and corresponding price appreciation.

Read: The average target price is 1.329 times the pre-announcement stock price, and the median is 1.255. Target price revisions are larger for upgrades and negative for many downgrades.

Economic message: Analysts' targets embed substantial expected upside. The target-to-price ratio is therefore naturally interpreted as an ex ante expected return measure, which becomes central in the cointegration analysis.

10.3 Figure 1: Average Buy-and-Hold Abnormal Return Around Announcements of Target Prices and Figure 2: IBM's Weekly Stock Price, Target Price, and Target Price-to-Price Ratio

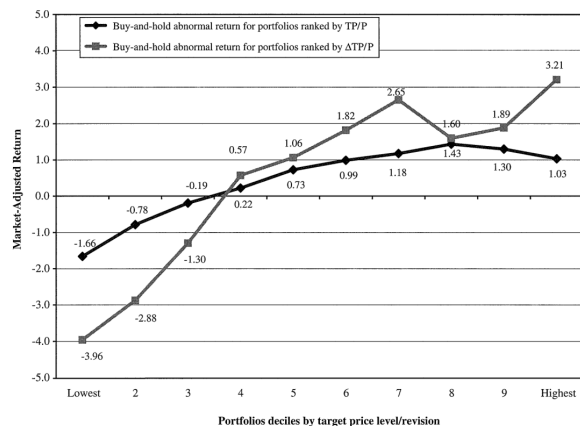


Figure 1. Average buy-and-hold abnormal return around announcements of target prices. This figure depicts average buy-and-hold abnormal returns for the period from two days prior to through two days subsequent to the announcement of target price for decile portfolios ranked on the basis of two target price information measures. Abnormal returns are computed as the difference between the firm buy and hold return and the buy and hold return on the NYSE/AMEX/Nasdaq value weight index. The information content measures are (a) the ratio of target price to preannouncement stock price (stock price outstanding two days prior to the announcement of the target price), denoted (TP/P), and (b) the change in the individual brokerage house target price scaled by preannouncement stock price, denoted ($\Delta TP/P$). To avoid a possible cross-correlation problem caused by identical return observations, we delete all but one of identical return observations within each portfolio.

Figure 1: Event-window abnormal returns by target price level/revision decile.

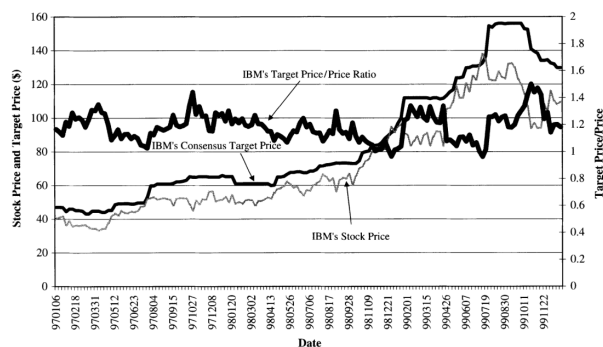


Figure 2. IBM's weekly stock price, target price, and target price-to-price ratio. This figure depicts weekly market price and consensus target price for IBM. The market prices are from CRSP and the weekly consensus target price is the average of all outstanding target prices issued over the preceding 90 days. All target prices are from the First Call database. We also plot the ratio of the target price to market price at each point in time, denoted by TP/P . The vertical axis on the left-hand side corresponds to the price series while the axis on the right-hand side corresponds to the TP/P ratio.

Figure 2: IBM weekly market price, consensus target price, and TP/P ratio.

Read: Figure 1 shows a clear monotonic relation between target price favorableness and event returns. Figure 2 shows that IBM's target price and market price move together over time, while their ratio fluctuates around a relatively stable level.

Economic message: The short-run figure says that target price revisions move stock prices. The long-run figure motivates the cointegration design: target and market prices are tied together but target prices are consistently above market prices because they are forward looking.

10.4 Table III: Relative Informativeness of Analyst Target Price, Stock Recommendation, and Earnings Forecast Revisions

Table III
Relative Informativeness of Analyst Target Price, Stock Recommendation, and Earnings Forecast Revisions
The sample consists of all target price announcements between January 1997 and December 1999. The table reports regression results in which the dependent variable is the market-adjusted buy-and-hold abnormal return around target price announcements and the independent variables are indicator variables for analyst recommendations, target prices, and earnings forecast revisions. The recommendation indicator variables assume the value of 1 for the relevant recommendation revisions and 0 otherwise. The recommendation categories are upgrades, downgrades, and reiterations. We also report p -values of tests of equality of the regression coefficients. Abnormal return is computed as the difference between the firm buy-and-hold return and the buy-and-hold return on the NYSE/AMEX/Nasdaq value-weight index over the period beginning two days prior to and ending two days subsequent to the target price announcement. Earnings forecast revision is computed as the percentage change in the brokerage house current and prior annual earnings forecast scaled by the preannouncement stock price. Target price revision is measured as the change in the brokerage house target price, scaled by the preannouncement stock price. We winsorize the earnings forecast and target price revision variables at the 1st and 99th percentiles to mitigate the possible effect of extreme observations and verify that regression results are not sensitive to influential observations.

Variable	Direction of Recommendation Revision				Sign of Target Price Revision		
	All Observations (1)	Upgrades (2)	Downgrades (3)	Reiterations (4)	Positive (5)	Negative (6)	Zero (7)
α_1 (Recommendation upgrade indicator)	2.966	2.702			4.311	0.919	1.210
	21.3	18.6			25.7	2.7	3.7
α_2 (Recommendation downgrade indicator)	-3.001		-2.493		0.048	-6.357	-1.799
	-20.5		-15.9		0.2	-26.8	-5.9
α_3 (Recommendation reiteration indicator)	0.375			0.376	1.571	-1.510	0.132
	12.1			12.1	29.0	-16.2	2.7
β (Earnings forecast revision)	1.003	1.078	0.967	1.093	1.008	0.902	1.003
	30.8	4.8	7.9	28.7	12.7	17.4	15.8
γ (Price target revision)	3.705	6.278	8.084	3.330	1.787	0.678	—
	38.7	14.8	18.1	33.1	12.9	3.1	—
Adjusted R^2	5.5%	6.6%	13.2%	3.2%	9.5%	11.5%	11.1%
N	70,852	3,176	2,866	64,780	27,561	17,239	26,052

(continued)

Table III, regression coefficients.

Table III (Continued)			
p -values of Tests of Equality of Regression Coefficients			
	All Observations (Column 1)	Recommendation Revision (Columns 2-4)	Target Price Revision (Columns 5-7)
$\alpha_1 = \alpha_2 = \alpha_3$	0.000		
$\beta(\text{Upgrade}) = \beta(\text{Downgrade}) = \beta(\text{Reiteration})$		0.619	
$\beta(\text{Upgrade}) = \beta(\text{Reiteration})$		0.948	
$\beta(\text{Downgrade}) = \beta(\text{Reiteration})$		0.327	
$\gamma(\text{Upgrade}) = \gamma(\text{Downgrade}) = \gamma(\text{Reiteration})$		0.000	
$\gamma(\text{Upgrade}) = \gamma(\text{Reiteration})$		0.000	
$\gamma(\text{Downgrade}) = \gamma(\text{Reiteration})$		0.000	
$\alpha_1(\text{Positive}) = \alpha_1(\text{Negative}) = \alpha_1(\text{Zero})$			0.000
$\alpha_2(\text{Positive}) = \alpha_2(\text{Negative}) = \alpha_2(\text{Zero})$			0.000
$\alpha_3(\text{Positive}) = \alpha_3(\text{Negative}) = \alpha_3(\text{Zero})$			0.000
$\beta(\text{Positive}) = \beta(\text{Negative}) = \beta(\text{Zero})$			0.382
$\gamma(\text{Positive}) = \gamma(\text{Negative}) = \gamma(\text{Zero})$			0.001

Table III, tests of equality of regression coefficients.

Read: The target price revision coefficient is highly significant in the full sample. Its magnitude differs across recommendation and target price revision states.

Economic message: Target prices are incrementally informative. The market uses recommendations, earnings forecasts, and target price revisions jointly rather than treating any one signal as sufficient.

10.5 Table IV: Postevent Cumulative Abnormal Returns and Table V: Calendar Time Regressions

Table IV

Postevent Cumulative Abnormal Returns

Size and book-to-market adjusted cumulative abnormal returns (CARs) are calculated for each event in our sample as follows. First, we obtain the market capitalization and book-to-market ratio for each firm prior to the event. Then, each firm is matched with a benchmark portfolio return from the Fama and French 25 size and book-to-market sorted portfolios. Third, a t -month CAR ($t = 1, \dots, 6$) is calculated by cumulating the event firm return beginning in the first month after the event through event-month + t minus the cumulative matching portfolio return over the same time period. We present CARs for subsamples of events classified by recommendation upgrades, downgrades, and reiterations. Within each recommendation upgrade and downgrade groups, we present CARs for tercile portfolios sorted based on the magnitude of the analyst's target price revision at the time of the event. For recommendation reiterations, we present CARs for decile portfolios sorted based on the magnitude of the analyst's target price revision at the time of the event. We winsorize monthly return observations at the 2nd and 98th percentiles to mitigate the possible effect of extreme observations. To avoid a possible cross-correlation problem caused by identical return observations, we delete all but one of identical return observations within each portfolio. Standard errors for the CAR estimates are obtained using the sample standard deviation of the abnormal returns. For example, inferences regarding the six-month CAR are based on the cross-sectional standard deviation of the events-firms' six-month CARs. The resulting t -statistics are presented below the CAR estimates. Within each possible recommendation/target price classification we also report the average of the target price revisions (scaled by preannouncement stock price). For example, for all events in which recommendations were upgraded, the average target price revision was 10 percent.

		Postevent Month					
	Average Target Price Revision	+1	+2	+3	+4	+5	+6
<u>Recommendation upgrades</u>							
All target price revisions	10%	1.03 4.1	1.45 4.0	1.73 3.9	2.66 5.1	2.82 4.8	3.08 4.7
Most favorable target price revisions	37%	1.97 4.2	2.83 4.1	3.21 3.7	4.23 4.2	4.98 4.4	5.21 4.2
Least favorable target price revisions	-20%	0.08 0.2	0.08 0.1	-0.23 -0.3	-0.20 -0.2	-0.13 -0.1	-0.38 -0.4
<u>Recommendation reiterations</u>							
All target price revisions	-1%	0.31 3.7	0.73 6.1	0.96 6.3	1.09 6.1	0.90 4.5	1.08 5.0
Most favorable target price revisions	32%	1.19 5.8	3.49 10.9	4.22 10.4	4.77 10.1	5.24 10.0	6.22 11.0
Least favorable target price revisions	-58%	-0.68 -4.3	-1.07 -4.5	-1.10 -3.9	-1.16 -3.5	-1.74 -4.7	-1.88 -4.6
<u>Recommendation downgrades</u>							
All target price revisions	-31%	-0.80 -3.0	-0.50 -1.3	-0.41 -0.8	-0.17 -0.3	-0.53 -0.9	-0.36 -0.5
Most favorable target price revisions	13%	-1.14 -2.7	-0.90 -1.5	-0.66 -0.9	-0.86 -1.0	-0.11 -0.1	0.13 0.1
Least favorable target price revisions	-91%	-0.63 -1.2	-0.40 -0.5	-0.08 -0.1	-0.23 -0.2	-1.43 -1.2	-1.19 -0.9

Table IV: Postevent cumulative abnormal returns.

Table V
Calendar Time Regressions

The sample is all target price announcements between January 1997 and December 1999. Portfolios are formed by including all events that were announced within the previous six months. The portfolios' equally weighted monthly returns, in excess of the risk-free rate, are regressed on the following four factors: *RMRF*, the excess return on the value-weighted market portfolio; *SMB*, the return on a zero investment portfolio formed by subtracting the return on a large firm portfolio from the return on a small firm portfolio; *HML*, the return on a portfolio of high book-to-market stocks less the return on a portfolio of low book-to-market stocks; and *PR12*, formed by taking the return on high momentum stocks minus the return on low momentum stocks. We report regression results for portfolios classified by recommendation upgrades, reiterations, and downgrades. We form tercile (decile) portfolios for recommendation upgrades/downgrades (reiterations) based on the magnitude of the analyst target price revision, which we then regress on the four factors. For example, conditional on a recommendation upgrade, we construct a portfolio that includes firms whose target price revision occurred within the previous six months and was in the top or bottom tercile at the time it was announced. We winsorize monthly return observations at the 2nd and 98th percentiles to mitigate the possible effect of extreme observations. Adjusted R^2 and t -statistics are presented for each regression.

	Intercept	<i>RMRF</i>	<i>SMB</i>	<i>HML</i>	<i>PR12</i>	Adjusted R^2
<u>Recommendation upgrades</u>						
All target price revisions	0.373 1.69	1.150 23.37	0.537 8.86	0.171 2.56	0.005 0.09	95.5%
Most favorable target price revisions	0.799 2.59	1.170 17.01	0.748 8.83	-0.169 -1.81	-0.022 -0.32	95.1%
Least favorable target price revisions	-0.112 -0.31	1.231 15.54	0.429 4.42	0.344 3.22	-0.061 -0.76	88.0%
<u>Recommendation reiterations</u>						
All target price revisions	0.094 0.70	1.129 38.62	0.686 18.86	0.221 5.34	-0.110 -3.30	98.7%
Most favorable target price revisions	0.478 3.07	1.161 33.07	0.780 17.33	-0.152 -3.08	-0.009 -0.28	99.0%
Least favorable target price revisions	-0.080 -0.31	1.232 22.26	0.690 9.48	0.216 2.74	-0.311 -4.87	95.4%
<u>Recommendation downgrades</u>						
All target price revisions	-0.325 -1.32	1.179 21.44	0.471 6.95	0.275 3.69	-0.132 -2.35	93.7%
Most favorable target price revisions	-0.297 -0.99	1.141 17.16	0.442 5.36	0.244 2.71	0.103 1.48	91.1%
Least favorable target price revisions	-0.354 -0.94	1.332 15.86	0.614 5.94	0.219 1.93	-0.538 -6.29	90.7%

Table V: Calendar-time four-factor regressions.

Read: Table IV shows large positive six-month drift after the most favorable target price revisions, especially for recommendation upgrades and reiterations. Table V confirms this pattern using calendar-time regressions.

Economic message: The market does not fully impound target price information immediately. Target price revisions have predictive content for subsequent returns, especially when the recommendation itself does not change.

10.6 Table VI: Cointegration Regression Results and Table VII: Informativeness of Target Prices Based on the Cointegration Regression

Table VI
Cointegration Regression Results

The table provides regression results of the cointegration analysis. Panel A presents results for IBM's 151 weekly observations of target and market prices spanning the period January 1997 through December 1999. The target price series is formed by equal-weighting all outstanding target prices that were issued within the previous 90 days. The regression setup is given in Section III and we estimate the parameter b , which captures the long-run ratio of target prices-to-market prices, and the vector α , which contains the response coefficients of each price variable to deviations from the long-run equilibrium. The first row contains the parameter estimates and the second contains the corresponding p -values. We do not report standard errors associated with b , as these are based on asymptotic theory whose finite sample properties are undetermined. Panel B provides results for the full sample of 900 firms that have at least one year of continuous record of weekly target and market prices. For each parameter we calculate the mean, median, 25th, 50th, and 75th percentile, as well as the standard deviation across the 900 firm regression estimate. Panel C provides regression results for subsamples of firms sorted by market capitalization (size). Size terciles are formed on the basis of NYSE capitalization cutoffs and are adjusted quarterly. For each size sort, we report the average of the regression estimates. For example, for the smallest 300 firms in this sample the average long-run ratio b is 1.37.

	Long-run Ratio b	Target Prices Response α_{tp} (%)	Market Prices Response α_{market} (%)	
Panel A: IBM				
	1.23	0.17	0.04	
	—	0.00	0.35	
Panel B: Full Sample				
Mean	1.28	0.09	− 0.02	
25th percentile	1.20	0.07	− 0.04	
Median	1.26	0.09	− 0.01	
75th percentile	1.33	0.11	0.01	
Standard deviation	0.11	0.04	0.04	
Panel C: Size-sorted Results				
Portfolio	Number of Firms	Long-run Ratio b	Target Prices Response α_{tp} (%)	Market Prices Response α_{market} (%)
Small	126	1.37	0.09	− 0.02
Medium	347	1.29	0.09	− 0.02
Large	427	1.23	0.10	− 0.01

Table VI: Long-run target-to-price ratios and response coefficients.

Read: Table VI estimates an average long-run target-to-market price ratio of 1.28 and shows that target prices do most of the error correction. Table VII shows that unexpected target price revisions, not expected revisions, drive event returns.

Economic message: The cointegration model gives a disciplined way to separate predictable analyst target revisions from genuine target price news. Investors appear to react primarily to the surprise component.

Table VII

Informativeness of Target Prices Based on the Cointegration Regression

The sample contains all target price announcements between January 1997 and December 1999. The table reports regression results in which the dependent variable is the market-adjusted buy-and-hold abnormal returns around target price announcements. The independent variables are indicator variables for analysts' recommendation revisions, earnings forecast revisions, and expected and unexpected target price revisions. The indicator variables assume the value 1 for the relevant recommendation revision and 0 otherwise. The recommendation categories are upgrades, downgrades, and reiterations. Abnormal returns are computed as the difference between the firm buy-and-hold return and the buy-and-hold return on the NYSE/AMEX/Nasdaq value-weighted index over the period beginning two days prior to and ending two days subsequent to the target price announcement. Earnings forecast revision, denoted $\Delta F/P$, is computed as the percentage change in the brokerage house current and prior annual earnings forecast scaled by preannouncement stock price. Expected target price revisions for each firm and event are constructed as follows. We first estimate a one-week-ahead forecast of the consensus target prices using the sample information that would have been available to investors prior to the release of the analyst report. We require a minimum of 10 weekly observations to fit the cointegration regression. The difference between the regression forecast and the preannouncement consensus target price, denoted (Expected $\Delta TP/P$), serves as a proxy for the expected consensus target price revision. Unexpected target price revision, denoted (Unexpected $\Delta TP/P$), is the difference between the announced and expected target price. Both the expected and unexpected target price variables are scaled by the event-firm's market price two days prior to the event. We winsorize all variables at the 1st and 99th percentiles to mitigate the possible effect of extreme observations. In addition, we verify that regression results are not sensitive to influential observations. The number of observations in each regression is 43,660.

Variable	Model I	Model II
z_1 (Recommendation upgrades)	2.120	2.171
	17.8	18.5
z_2 (Recommendation downgrades)	− 2.188	− 1.887
	− 14.0	− 12.9
z_3 (Recommendation reiterations)	− 0.070	0.162
	1.9	3.6
β ($\Delta F/P$)	2.153	2.173
	29.4	30.7
γ (Expected $\Delta TP/P$)	2.326	
	1.8	
η (Unexpected $\Delta TP/P$)	2.671	
	27.0	
η_1 (Unexpected $\Delta TP/P$ above/away)		2.712
		17.7
η_2 (Unexpected $\Delta TP/P$ above/towards)		1.288
		1.7
η_3 (Unexpected $\Delta TP/P$ below/away)		6.864
		20.1
η_4 (Unexpected $\Delta TP/P$ below/towards)		1.494
		9.6
R^2	5.9%	6.7%
p -values of tests of equality of coefficients		
$\eta_1 = \eta_2$		0.072
$\eta_3 = \eta_4$		0.000
$\eta_1 = \eta_3$		0.000
$\eta_2 = \eta_4$		0.794

Table VII: Event returns using expected and unexpected target price revisions.

10.7 Table A1: The Role of Recommendation Revisions

Table A1 The Role of Recommendation Revisions						
Variable	Recommendation Upgrades from			Recommendation Downgrades from		
	Hold to Strong Buy (1)	Buy to Strong Buy (2)	Hold to Buy (3)	Strong Buy to Hold (4)	Strong Buy to Buy (5)	Buy to Hold (6)
α (Intercept)	3.459	2.479	2.670	- 3.239	- 1.960	- 2.655
	9.3	13.1	9.3	8.6	9.0	9.2
β (Earnings forecast revision)	0.497	1.051	1.260	0.260	2.443	0.813
	1.0	3.1	3.3	1.4	9.5	3.7
γ (Target price revision)	5.639	7.779	4.605	9.093	6.798	8.600
	5.9	12.3	6.4	10.1	10.8	9.4
Adj. R^2	5.7	7.9	5.8	14.6	14.3	12.3
N	517	1,879	780	554	1,465	887

Table A1: The role of recommendation revisions.

Read: Larger recommendation moves have stronger announcement effects even after controlling for earnings forecast and target price revisions. For example, a hold-to-strong-buy upgrade produces a larger intercept than a buy-to-strong-buy upgrade.

Economic message: Recommendation magnitude conveys confidence or precision. A target price may summarize the mean valuation view, while the recommendation change helps investors infer how strongly the analyst believes the valuation signal.

11 Short summary

- Target prices are informative in the short run: abnormal returns around announcements increase with target price favorableness.
- Target price revisions are not subsumed by recommendations or earnings forecasts.
- Target prices matter most when recommendation and target price signals reinforce each other.
- Recommendation reiterations can still be informative if the target price changes.
- Favorable target price revisions predict positive post-event abnormal returns.
- Consensus target prices and market prices share a long-run relation; on average, target prices are 28% above market prices.
- Analysts adjust target prices toward the long-run relation much more than market prices adjust.
- Investors react to unexpected target price revisions, which suggests they understand at least part of the predictable target-price adjustment process.

12 Conclusion

Brav and Lehavy (2003) show that analyst target prices are economically meaningful signals. They communicate valuation magnitude beyond the information in recommendation labels and earnings forecast revisions. The paper's strongest insight is that target prices operate at two frequencies. In the short run, target price revisions move stock prices and predict subsequent abnormal returns. In the long run, target prices and market prices are tied by a stable relation in which target prices sit above market prices because they are forward-looking.

The paper also changes how one should read analyst reports. A recommendation alone is incomplete: a buy with a small target price change is very different from a buy with a large target price increase. Likewise, a hold recommendation with a large target revision may contain meaningful information even without a recommendation change. The economic role of target prices is therefore to reveal the analyst's valuation intensity.